ISHRAQ TASHDID

Hardware Security • AI-Assisted Verification • Formal Methods • Secure Semiconductor Design

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Research Interests

Security assurance for advanced semiconductor systems; AI-assisted hardware vulnerability analysis and threat modeling; formal and property-based security verification; security-aware electronic design automation; security knowledge preservation across synthesis and implementation; hardware trust and authentication; secure heterogeneous integration and chiplet-based systems; secure test and design-for-test; trustworthy SoC and accelerator design.

Google Scholar (Oct. 2026): **35 citations** | h-index: 3 | i10-index: 1.

Education

University of Central Florida (UCF), Orlando, FL, USA *Jan. 2024–Present*
Ph.D. in Computer Engineering **Expected Jan. 2028**
M.S. in Computer Engineering **Conferred Aug. 1, 2026**
GPA: **3.907/4.00** | Advisor: Dr. Sazadur Rahman, Department of Electrical and Computer Engineering
UCF Trustee's Doctoral Fellow | **Future Faculty Laureate**

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh *Feb. 2017–Jun. 2022*
B.Sc. in Electrical and Electronic Engineering CGPA: 3.48/4.00

Research Experience

Graduate Research Assistant — University of Central Florida, Orlando, FL *Jan. 2024–Present*
Hardware Security and Trustworthy Computing Lab; advised by Dr. Sazadur Rahman

- **AI-assisted hardware security verification and vulnerability reasoning.** Developed methods connecting standardized vulnerability knowledge, hardware structural information, AI-assisted reasoning, and formal/property-based verification to automate security analysis of semiconductor designs. This research includes ATLAS and continuing work on Aegis and related security-verification frameworks.
- **ATLAS — threat-to-assertion learning for SoC security.** Developed an AI-assisted framework that identifies hardware assets, maps relevant weaknesses from standardized sources such as CWE, constructs structured threat models, and generates assertion-based security properties and formal-verification scripts. Across three HACK@DAC benchmark designs, ATLAS identified 39 of 48 evaluated weaknesses and generated correct security properties for 33 of the 39 identified cases.
- **Security knowledge across synthesis and implementation.** Investigating how security assets, assumptions, countermeasures, verification obligations, and policies can be preserved and checked across synthesis transformations. Secure-SDC focuses on retaining security knowledge and verification obligations across synthesis.
- **Security assurance for open-source semiconductor design flows.** Since Sept. 2026, contributing to SecureOpenROAD research on security assurance across synthesis and physical implementation. Identified security-critical metrics for semiconductor design flows that helped shape the methodology in the submitted research proposal.
- **Security of heterogeneous and chiplet-based systems.** Developed authentication and trust architectures for System-in-Package and chiplet-based systems, including SAFE-SiP, AuthenTree, InterPUF, and RoutePUF. The work combines multi-party computation, physically unclonable functions, distributed trust, and interposer-resident security mechanisms.
- **Secure test and verification for heterogeneous systems.** Investigating security-aware test and verification

for chiplet-based systems, including protection of authentication, test, and inter-chiplet information in multi-vendor environments.

- **Hardware IP protection and reconfigurable security.** Developed ECOLogic, an eFPGA-augmented SoC methodology supporting post-fabrication adaptability, logic obfuscation, and hardware IP protection while maintaining practical implementation characteristics.
- **Machine learning for semiconductor physical design.** Developed BeyondPPA, a human-inspired reinforcement-learning methodology for macro placement that incorporates practical physical-design considerations and post-route reliability objectives beyond conventional PPA optimization. The work was selected as a Best Paper Award Nominee at MLCAD 2025.

Industry Experience

XPU Security Research Intern — Intel Corporation, Folsom, CA, USA *May–Sep. 2026*

- Conducted security research for next-generation XPU/accelerator architectures within Intel's hardware-security research environment.
- Applied AI-assisted security analysis, hardware vulnerability reasoning, and rigorous verification techniques to large-scale hardware-security problems.
- Collaborated with researchers and engineers working across AI, hardware security, processor/accelerator architecture, and verification.
- Supervisor: James Cavanaugh, Intel Corporation.

Hardware Design & Verification Engineer — Dynamic Solution Innovators Ltd., *Jan. 2022–Dec. 2023*
Bangladesh
Client project: XCelerium, Irvine, California

- Performed design and verification work for custom RISC-V processor implementations using SystemVerilog, UVM, and cocotb.
- Verified processor functionality at ISA and system level, including floating-point functionality and processor subsystems.
- Worked on APB-to-AXI bridges and AXI memory-controller verification, including protocol behavior, assertions, corner cases, and coverage.
- Developed Python-based verification automation for test execution, regression analysis, coverage convergence, and debugging.

Peer-Reviewed Publications

1. **Ishraq Tashdid**, Kimia Tasnia, Alexander Garcia, Jonathan Valamehr, and Sazadur Rahman, "ATLAS: AI-Assisted Threat-to-Assertion Learning for System-on-Chip Security Verification," *63rd ACM/IEEE Design Automation Conference (DAC)*, Long Beach, CA, USA, 2026.
   doi:10.1145/3770743.3804400
2. **Ishraq Tashdid**, Tasnuva Farheen, Navid Asadi, and Sazadur Rahman, "RoutePUF: An Interposer-Resident Delay PUF with MPC for Zero-Trust Chiplet SiPs," *Government Microcircuit Applications and Critical Technology Conference (GOMACTech)*, 2026.
3. **Ishraq Tashdid**, Tasnuva Farheen, and Sazadur Rahman, "InterPUF: Distributed Authentication via Physically Unclonable Functions and Multi-party Computation for Reconfigurable Interposers," *IEEE International Symposium on Hardware Oriented Security and Trust (HOST)*, 2026, pp. 244–255.
   doi:10.1109/HOST68814.2026.11604869
4. **Ishraq Tashdid**, Dewan Saiham, Nafisa Anjum, Tasnuva Farheen, and Sazadur Rahman, "ECOLogic: Enabling Circular, Obfuscated, and Adaptive Logic via eFPGA-Augmented SoCs," *43rd IEEE International Conference on Computer Design (ICCD)*, 2025, pp. 523–527.

doi:10.1109/ICCD65941.2025.00080

5. **Ishraq Tashdid**, Tasnuva Farheen, and Sazadur Rahman, "SAFE-SiP: Secure Authentication Framework for System-in-Package Using Multi-party Computation," *Proceedings of the Great Lakes Symposium on VLSI (GLSVLSI)*, 2025, pp. 391–396.
doi:10.1145/3716368.3735248

6. **Ishraq Tashdid**, Tasnuva Farheen, and Sazadur Rahman, "AuthenTree: A Scalable MPC-Based Distributed Trust Architecture for Chiplet-Based Heterogeneous Systems," *IEEE Physical Assurance and Inspection of Electronics (PAINE)*, 2025.
doi:10.1109/PAINE66113.2025.11320196

7. **Ishraq Tashdid**, Valentina Terry, Jordan Merkel, Tasnuva Farheen, and Sazadur Rahman, "BeyondPPA: Human-Inspired Reinforcement Learning for Post-Route Reliability-Aware Macro Placement," *7th ACM/IEEE Symposium on Machine Learning for CAD (MLCAD)*, Santa Cruz, CA, USA, 2025, pp. 1–12.
Best Paper Award Nominee. doi:10.1109/MLCAD65511.2025.11189164

Ongoing Research and Manuscripts in Preparation

- **Aegis: AI-Assisted and Evidence-Grounded Security Evaluation of Hardware Designs.** Apr. 2026–Present. Developing scalable hardware-vulnerability identification and validation using AI-assisted reasoning, structural analysis, formal verification, directed simulation, and coverage-driven testing. Manuscript in preparation.
- **PROTON.** Ongoing research in AI-assisted and rigorous hardware-security verification; manuscript in preparation.
- **Secure-SDC / Security Knowledge Across Synthesis.** Ongoing research on preserving and validating security knowledge, assets, policies, assumptions, countermeasures, and verification obligations across synthesis transformations; manuscript in preparation.
- **SecureOpenROAD: Security Assurance for Open-Source Semiconductor Design and Physical Implementation.** Sept. 2026–Present. Ongoing work on security-aware open-source semiconductor implementation, including evaluation of security intent across synthesis and physical-design stages.
- **Security-Aware Testing and Verification for Heterogeneous Semiconductor Systems.** Research on secure test, authentication, and verification mechanisms for heterogeneous and chiplet-based systems; manuscript in preparation.

Published Book Chapter

Ishraq Tashdid, Kimia Tasnia, and Sazadur Rahman, "Introduction to Large Language Models (LLMs)," in *LLM for SoC Design and Security*, Springer, Cham, pp. 31–59, 2027 (first published online Oct. 16, 2026).
doi:10.1007/978-3-032-34116-7_2

Patents & Inventions

U.S. Provisional Patent Application — University-Authorized Filing *2026*
"Systems and Methods for AI-Assisted Threat-to-Assertion Generation in Hardware Security Verification"
Inventors: Sazadur Rahman, **Ishraq Tashdid**, and Kimia Tasnia.
UCF Office of Technology Transfer reference: **2026-163-01**.
UCF engaged patent counsel to prepare the provisional application, reviewed its draft, and authorized filing on October 22, 2026.

Research Proposal Contributions

Proposal Contributor *2026*
DOE Genesis Mission: Transforming Science and Energy with AI — Research Proposal
PI: Dr. Sazadur Rahman; collaboration with Silvaco Inc.

- Contributed technical research content on AI-assisted semiconductor-security analysis, microelectronics, and rigorous verification.

Proposal Contributor *2026*
ASCEND — Security and Trust for Heterogeneous Chiplet Systems
PI: Dr. Tasnuva Farheen (LSU); Co-PI: Dr. Sazadur Rahman (UCF); NSF SaTC research proposal.

- Contributed research on chiplet authentication, distributed trust, interposer security, and heterogeneous packaging assurance.

Proposal Contributor *2025–2026*
SecureOpenROAD — Security Assurance for Open-Source Semiconductor Design Flows
PI: Dr. Sazadur Rahman; collaborators: Prof. Andrew B. Kahng, Kimia Azar, Hadi Kamali, and Precision Innovations Inc.

- Developed security-critical evaluation metrics that informed the proposed methodology for measuring security during semiconductor synthesis and physical implementation.

Invited Talks & Presentations

Intel AI Forum — Intel Corporation *Feb. 2026*
Invited talk to Intel researchers on ATLAS and AI-assisted formal security verification.

4th Annual Florida Semiconductor Summit — Orlando, FL *Feb. 2026*
Invited presentation on hardware security and AI-assisted verification within the semiconductor ecosystem.

DAC 2026 Open-Source EDA Birds-of-a-Feather — Long Beach, CA *Jul. 2026*
Invited rapid-fire presentation, *"Security Challenges and Opportunities in the Open-Source EDA Ecosystem."*

IEEE UCF Student Branch General Body Meeting — Orlando, FL *Oct. 2026*
Invited speaker on formal verification, hardware security, and the transition from academic research to industry.

Teaching & Mentoring

Graduate Teaching Assistant / Grader — UCF Department of Electrical and Computer Engineering
Spring 2024; Summer 2025; Fall 2025

- Supported courses including Engineering Analysis and Computation, Linear Systems II, Electronics II Laboratory, and Digital Systems Laboratory.
- Responsibilities included grading, office hours, laboratory instruction, student support, and assessment.

Undergraduate Research Mentor *Aug. 2024–Present*

- Mentored undergraduate researchers in semiconductor physical design, machine-learning-assisted EDA, and hardware research.
- Mentored students who coauthored the MLCAD 2025 Best Paper Award Nominee and subsequently obtained summer research internships at AMD and Northrop Grumman.

Professional Service

- **Program Committee Member**, Trustworthy Agentic Systems (TAS 2026), AAAI 2026 Fall Symposium Series.
- **External Reviewer**, ACM/IEEE Design Automation Conference (DAC), 2025 and 2026.
- **External Reviewer**, IEEE International Conference on Computer Design (ICCD), 2025.
- **External Reviewer**, IEEE International Symposium on Hardware Oriented Security and Trust (HOST), 2026.
- **Invited Manuscript Reviewer**, *IEEE Transactions on Mobile Computing*, 2026.
- **Invited Reviewer**, *IEEE Design & Test*, 2026.

Honors, Awards & Fellowships

- **UCF Trustee's Doctoral Fellowship**, University of Central Florida, 2024–2028 — four-year doctoral fellowship, $25,000 per year.
- **UCF ECE Outstanding Student Researcher Award**, May 2026 — departmental recognition for research contributions during the 2025–2026 academic year.
- **DAC Young Fellow**, 63rd ACM/IEEE Design Automation Conference, 2026.
- **Future Faculty Laureate**, UCF Department of Electrical and Computer Engineering, 2024–Present.
- **Best Paper Award Nominee**, ACM/IEEE Symposium on Machine Learning for CAD (MLCAD), 2025, for *BeyondPPA*.
- **NSF Travel Grant**, IEEE International Symposium on Hardware Oriented Security and Trust (HOST), 2025.

Technical Skills

Security & Analysis	Hardware threat modeling; CWE/CVE/CAPEC-based vulnerability reasoning; AI-assisted security analysis; assertion-based verification; formal security verification; static analysis; hardware IP protection; chiplet and SiP security; PUF-based authentication; multi-party computation.
Formal Verification	Cadence JasperGold; SystemVerilog Assertions; property generation; assertion-based verification; formal-property validation.
Programming	Python; SystemVerilog; Verilog; C; MATLAB; RISC-V assembly; Tcl; Shell scripting.
Design Verification	SystemVerilog/UVM; cocotb; pytest; coverage-driven verification; RISC-V ISA and system-level verification; AXI/APB protocols; CDC verification.
EDA / ASIC Flow	Cadence Genus; Cadence Innovus; Cadence JasperGold; Synopsys Design Compiler; Synopsys VCS; Synopsys IC Compiler II; Synopsys PrimeTime; OpenROAD; Yosys; ModelSim; Quartus Prime; PyVerilog.
AI / Machine Learning	PyTorch; large language models; prompting and structured reasoning; retrieval-augmented generation; reinforcement learning; PPO; DDPG; AI-assisted design and verification.
Development	Linux; Git; GitHub; LaTeX; Python-based automation; research software development.